

SYSTEM ARCHITECTURE – TAP2MED

1. What we're building

The neutral infrastructure layer of Indian primary care. A passive care event capture network across 1.4M clinics that logs anonymous health journeys, enables patient-controlled data consent, and becomes the rails every health startup, pharma company, insurer, and government health scheme plugs into.

1 scan -> 2 taps. Complete primary care intelligence. Zero friction. Zero cost to clinics. Forever.

2. Core principles:

- Clinics are permanently FREE. No exceptions at any point in future.
 - Personal data never leaves the clinic. Architecturally enforced.
 - Patient holds the consent key. Not clinic. Not insurer. Not us.
 - We are the protocol. Never the product built on top.
 - We never help pharma influence prescribing. Only intelligence.
 - Neutrality is the moat. The day we compromise it, we die.
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3. The three layer identity system:

Layer 1 - Clinic layer

- a) Clinic knows hash x9f3 = Ramesh Kumar
- b) That mapping never leaves the clinic
- c) Clinic cannot see x9f3's behaviour outside their walls

Layer 2 - Network layer

- a) Tap2Med sees x9f3 visited clinic A, lab B, pharmacy C
- b) Rich longitudinal journey
- c) Zero idea whose journey it is

Layer 3 - Consent layer

- a) Patient says yes
- b) Journey unlocks for specific purpose
- c) Specific time period

- d) Revocable anytime
 - e) Patient holds the key always
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4. Hashing scheme:

Two tokens generated per event. Phone number lives in memory for milliseconds. Never stored anywhere.

Local token, unique per clinic

Clinic sees this, maps to their patient locally

local_token = HMAC_SHA256(phone, SECRET + clinic_salt)

Network token, consistent across all clinics

Only Tap2Med uses this internally

network_token = HMAC_SHA256(phone, SECRET)

Same patient. Different local token at every clinic. Same network token everywhere.
Cross-clinic journey tracking without identity exposure.

5. Patient check-in sequence

Step 1 - Patient walks in, scans QR on clinic wall

Step 2 - Lands on scan.html

URL contains clinic_id

tap2med.com/scan?clinic=abc123

Step 3 - Returning visitor - phone retrieved from cookie

First time visitor, single phone number input

Step 4 - POST /event/log called

phone + clinic_id sent over HTTPS

Step 5 - Backend generates both tokens immediately

phone discarded - never written anywhere

Step 6 - Event written to DB

local_token + network_token + timestamp only

Step 7 - local_token returned to clinic dashboard

network_token never leaves server

Step 8 - Cookie set on patient device

Stores local_token - not phone number

Auto check-in on return visit

Zero typing next time

Step 9 - Clinic dashboard updates

Doctor sees new check-in

Opens prescription notepad

6. Prescription flow:

Doctor opens prescription.html on phone

Clinic profile auto-filled from localStorage

- clinic name, doctor name, reg number, address
- set once on registration, never typed again

Doctor types drug name + dosage + instructions(optional)

- 3 fields only
- phone keyboard friendly

Hits Generate

- PDF created on device instantly
- clinic branding + digital signature
- Tap2Med watermark bottom
- jsPDF, zero server involvement

Hits Share

- native Web Share API opens
- WhatsApp / SMS / Telegram
- doctor's choice, doctor's cost
- Tap2Med never touches message
- cost to Tap2Med: ₹0

What reaches Tap2Med servers:

- drug_category (inferred, broad)
- inferred_symptom (from drug map)
- local_token
- network_token
- timestamp

What never reaches Tap2Med servers:

- patient name
- prescription PDF content
